National Artificial Intelligence Research Resource (NAIRR)

NAIRR Pilot Overview & Resources at SDSC

CIML Workshop, June 26, 2026

Mahidhar Tatineni

University of California, San Diego

Halıcıoğlu School of Data Science and Computing

San Diego Supercomputer Center

SAN DIEGO SUPERCOMPUTER CENTER

UC San Diego

Outline

- National Artificial Intelligence Research Resource (NAIRR)
  - Overview
  - Opportunities under NAIRR Pilot Program
- NAIRR Research Resources
- NAIRR Educational Resources
- NAIRR Resources at SDSC
  - Voyager
  - Expanse CPU & GPU
  - Expanse AI Resource
  - PNRP Classroom
- Steps to get access to resources under NAIRR Pilot Program

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NAIRR Pilot Program Overview

- NAIRR's goal is to connect U.S. researchers and educators to AI resources to help them conduct their research activities and to train the next generation
- Resources include computation, data, software, models, training, and educational materials
- World-class AI resources come from both public and private-sector providers
- Reference:

https://nairrpilot.org/about/overview

Opportunities under NAIRR Pilot

- Startup Project Requests
  - Small light-weight request
  - Reviewed in 2 weeks
  - Helps researchers test the NAIRR resources, port their codes, assess their needs and if needed prepare a large-scale research request
  - Maximum award duration is 3 months

- Research Requests
  - Full scale requests for computing, model, platform and educational resources for AI research
  - Open to US-based researchers and educators from US-based institutions including academic institutions, non-profits, federal agencies or federally funded R&D centers, state, local, or tribal agencies, startups and small businesses with federal grants.
  - Maximum award duration is 12 months

Opportunities under NAIRR Pilot

- NAIRR Pilot Classroom Requests
  - Open to US based educators and researchers who are teaching undergraduate or graduate courses or shorter duration training sessions
  - Students must be also located in the US
  - Courses from any discipline are eligible for this program.

- NAIRR Pilot Deep Partnerships

  https://nairrpilot.org/opportunities/deep-partnerships

- Open Data, Models, and More

  https://nairrpilot.org/pilotresources

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NAIRR Research Resources: Focus Areas

- Advancing AI methods that enable scientific discovery and improve AI interpretability, security and trust.
- Accelerating time to science and innovation through AI enabled automation, autonomy and novel design and control processes.
- Applying AI to use, share, or integrate sensitive data from multiple sources to enable new experimental methods and discovery.
- Advancing approaches for integrating simulations and AI.
- Creating or developing open-source AI tools, models, datasets, and methods.
- Training and educating the next generation AI-savvy workforce.

Other areas of AI research and education are accepted if they are in line with NAIRR Pilot objectives.

Reference: *https://nairrpilot.org/opportunities/allocations*

NAIRR Research Resources

- Amazon Web Services
- Anthropic Model API
- Cerebras Wafer-Scale Engine 2 (CS-2) AI Accelerator
- Chameleon
- CloudBank Research
- Databricks Data Intelligence Platform
- DOE Argonne National Laboratory AI Testbed
- EleutherAI Collaborations
- FABRIC
- Groq LPU Inference Engine
- Hugging Face Spaces and Compute Grants
- Indiana Jetstream2 GPU
- NCSA Delta GPU
- NCSA DeltaAI
- NVIDIA DGX Cloud
- OpenAI API
- Partnership to Advance Throughput Computing (PATh)
- PSC Bridges-2 Extreme Memory (PSC Bridges-2 EM)
- PSC Bridges-2 GPU (PSC Bridges-2 GPU)
- PSC Bridges-2 Regular Memory (PSC Bridges-2 RM)
- PSC Neocortex CS-2
- Purdue Anvil AI
- Purdue Anvil CPU
- Purdue Anvil GPU
- Sage Edge Computing Platform
- SambaCloud (Faster Inference - Mini-Max, DeepSeek, Maverick, more)
- SDSC Expanse AI Resource
- SDSC Expanse CPU
- SDSC Expanse GPU
- SDSC Voyager (Habana Training and Inference Processor based AI System)
- TACC Vista (NVIDIA GH100 Grace Hopper Superchip)
- TAMU ACES
- Voltage Park
- Weights & Biases MLOps Platform

Reference: https://nairrpilot.org/opportunities/allocations

NAIRR Research Resources

- Expand each resource tab for details on the offering
- CloudBank Research Example

CloudBank Research

Resource Type:	Cloud Platform
Organization:	CloudBank
Units:	Dollars
User Guide:	Link to User Guide
Features Available:	• Commercial Cloud • Innovative / Novel Compute • ACCESS Allocated • Globus Data Transfer • JupyterHub • AI tools and support • Model inference services • Research Collaboration • Federal agency systems • NAIRR Pilot • Active

Resource Description

CloudBank enables researchers to access commercial cloud platforms through the ACCESS and NAIRR Pilot allocation processes. Awards are made in dollars and usable across the following commercial cloud platforms: Amazon Web Services, Google Cloud, IBM Cloud, and Microsoft Azure. After onboarding, the CloudBank portal automatically creates cloud accounts for the platforms listed in your request, with the option to add more later. In the CloudBank Portal, you can manage which authorized users have permission to your cloud accounts and easily access the cloud vendor platforms using federated login with your ACCESS credentials. CloudBank continuously monitors resource usage, applies charges against your allocation, and delivers automated spend alerts. Regulated data is also supported on CloudBank.

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NAIRR Education Resources

- **CloudBank Classroom**
  - Cloud-hosted, small-scale JupyterHub designed for teaching, built on an open-source interactive computing stack (Python, R, JupyterLab, and related libraries)
  - Authentication leverages university identity management systems => campus credentials can be used
  - Preconfigured with widely used data science tools

- **Indiana Jetstream2 GPU**
  - Can run persistent education infrastructure services
  - Virtual machines for students
  - GPU resources (including vGPUs option) for AI/ML classes

- **NIH Cloud Lab**
  - Resources for instructors teaching courses and workshops related to life science
  - Tutorials available for bioinformatics, life science AI, and generative AI use cases
  - *Sensitive/restricted data *not* supported*

NAIRR Education Resources

- **Prototype National Research Platform (PNRP) classroom**
  - GPU resources for classes, workshops, tutorials, and hackathons that make use of AI/ML
  - Authentication based on InCommon – can leverage campus credentials
  - Kubernetes cluster to run the workloads, JupyterHub setup templates provided
  - PNRP staff and community supported
  - LLM services provided (running popular open weight models), can be integrated with Jupyter

- **Vocareum AI Notebook**
  - Cloud hosted Jupyer-based Notebook, customizable container images
  - Learning management system integration
  - Auto-graded assignments
  - Collaborative group work capabilities
  - Virtual Tutors
  - Can integrate with other resources on the backend, e.g. PNRP Nautilus cluster

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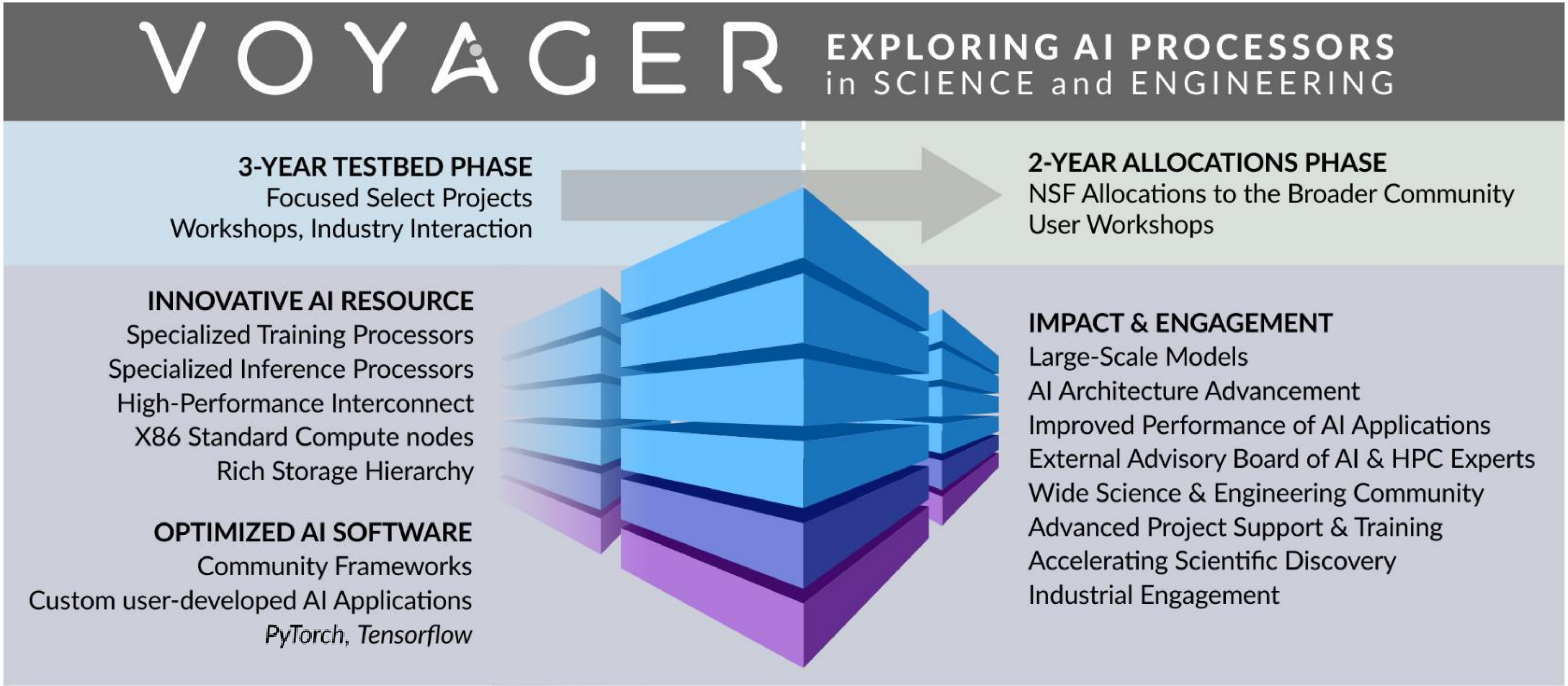
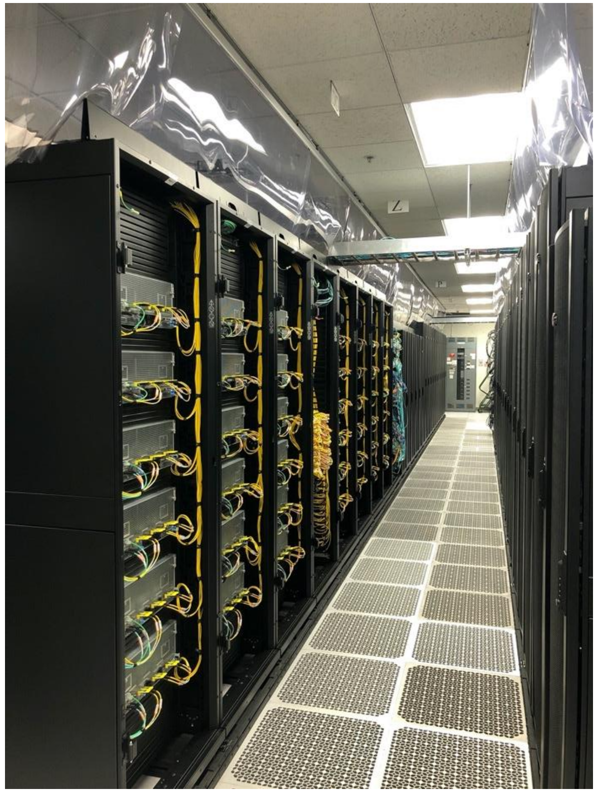
SDSC NSF ACCESS/NAIRR Allocated Resources

- SDSC operates and supports users on several NSF ACCESS/NAIRR allocated resources
- NSF resources are in two categories:
  - **Category I systems** – large-scale, production-ready computational resources that serve as capacity resources. At SDSC Expanse fits under this category
  - **Category II systems** - Innovative Prototypes/Testbeds for advanced computing and data analysis. These resources are experimental systems that deploy novel technologies, architectures, and usage models. At SDSC Voyager, PNRP, and Cosmos fit under this category.
- Category II systems typically have two phases: 1) the initial 3-year testbed phase where there are focused long term engagements with research groups to evaluate the new technologies and develop usage models, and 2) the 2-year allocated phase where systems are open to all users under the ACCESS or NAIRR Pilot programs
- Currently **Voyager and PNRP are in the allocations phase** and **Cosmos is in the testbed phase**



Category II System, NSF Award # 2005369

PI: Amit Majumdar (SDSC); Co PIs: Rommie Amaro (UCSD), Javier Duarte (UCSD), Mai Nguyen (SDSC), Robert Sinkovits (SDSC)

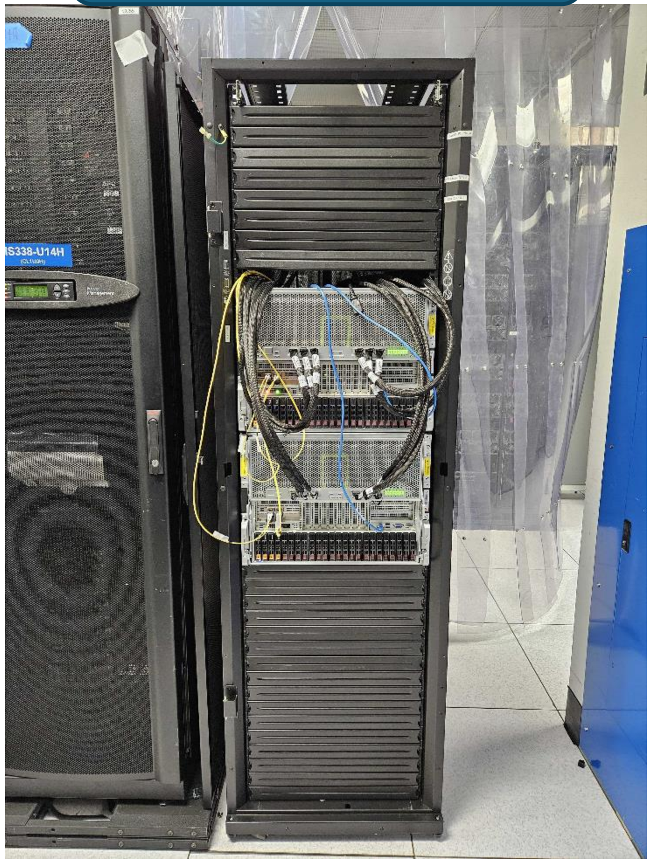


Voyager row at the SDSC Data Center

6x Gaudi training nodes/rack (7 racks total)

Arista 7808R 400 GbE switch

New since September 2024



Gaudi2 training nodes

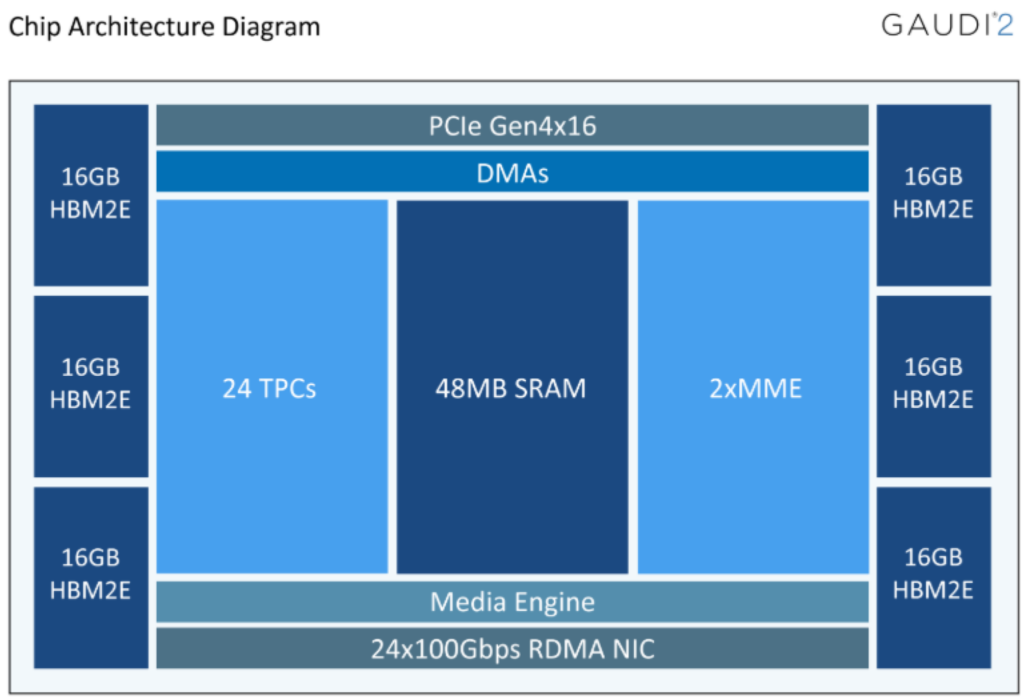
Voyager is a heterogeneous system designed to support complex AI workflows

- **42x Intel Habana Gaudi** training nodes, each with 8 training processors (**336 in total**); all-to-all network between processors on a node.
- Original configuration was with Gaudi1 processors. Most nodes switched to Gaudi2 processors this year.
- Gaudi processors feature specialized hardware units for AI, HBM2, and on-chip high-speed Ethernet
- **2x first generation inference nodes**, each with 8 inference processors (**16 in total**)
- **36x Intel x86 processors compute nodes** for general purpose computing and data processing
- **400 GbE interconnect** using RDMA over Converged Ethernet
- **3 PB Storage** system connected vis 25GbE. Deployed as Ceph; experiment with other filesystems
- 324 TB HFS; connectivity to compute via 25GbE
- Machine integrated by Supermicro; includes Arista switch

System Component	*Configuration*
INTEL GAUDI TRAINING NODES	
Node count	42
Training processors/node	8
Host x86 processors/node	2
Memory/node	512 GB DDR4
Memory/training processor	32 GB HBM2
Local NVMe	6.4 TB
INTEL GOYA INFERENCE NODES	
Node count	2
Inference processors/node	8
Host x86 processors/node	2
Memory/node	512 GB DDR4
Memory/inference processor	16 GB DDR4
Local NVMe	3.2 TB
STANDARD COMPUTE NODES	
Node count	36
x86 processors/node	2
Memory capacity	384 GB
Local NVMe	3.2 TB
STORAGE SYSTEM	
High performance storage: HDD:NVMe	3 PB:140 TB
High performance filesystems	Ceph, Lustre
Home filesystem storage: HDD:NVMe	324 TB: 12.4 TB
File system	NFS

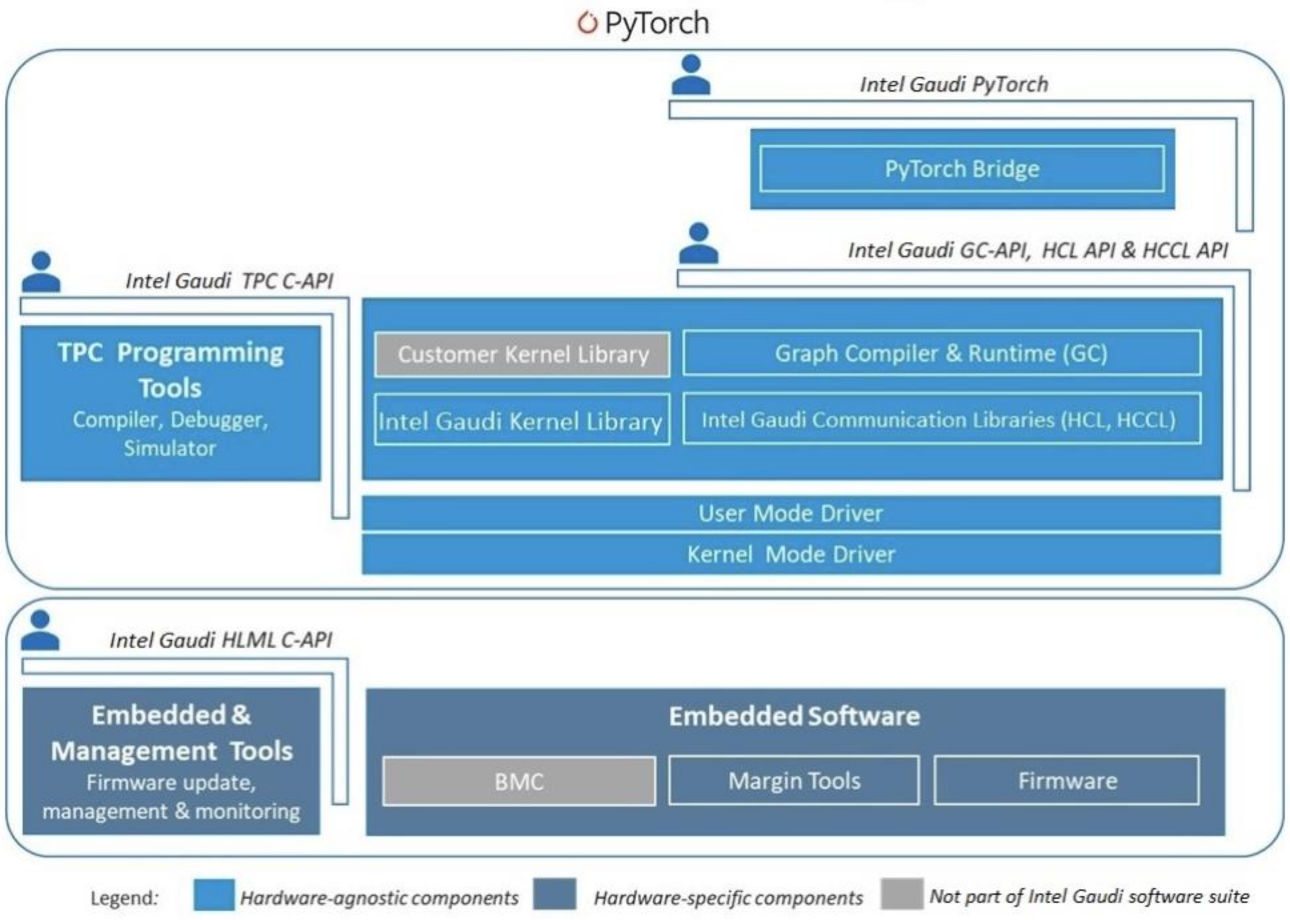
Gaudi2 Nodes

- Gaudi2 node
  - 7nm process technology
  - 24 Tensor Processor Cores (TPC)
  - Dual matrix multiplication engines (MME)
  - 24 100 Gigabit Ethernet integrated on chip
  - 96 GB HBM2E memory on board
  - 48 MB SRAM



https://docs.habana.ai/en/latest/Gaudi_Overview/Gaudi_Architecture.html#intel-gaudi-2-ai-accelerator

Intel Gaudi Software Suite for High Performance DL Training/Inference



Intel Gaudi Software Suite

- Shared software suite for training and inference
- Start running on Habana accelerators with minimal code changes
- Integrated with PyTorch. TensorFlow supported on older Synapse versions.
- Rich library of performance-optimized kernels
- Advanced users can write their custom kernels
- Components include graph compiler and runtime, TPC kernel library, firmware and drivers, and developer tools

Reference:
https://docs.habana.ai/en/latest/Gaudi_Overview/Intel_Gaudi_Software_Suite.html

EXPANSE

COMPUTING WITHOUT BOUNDARIES

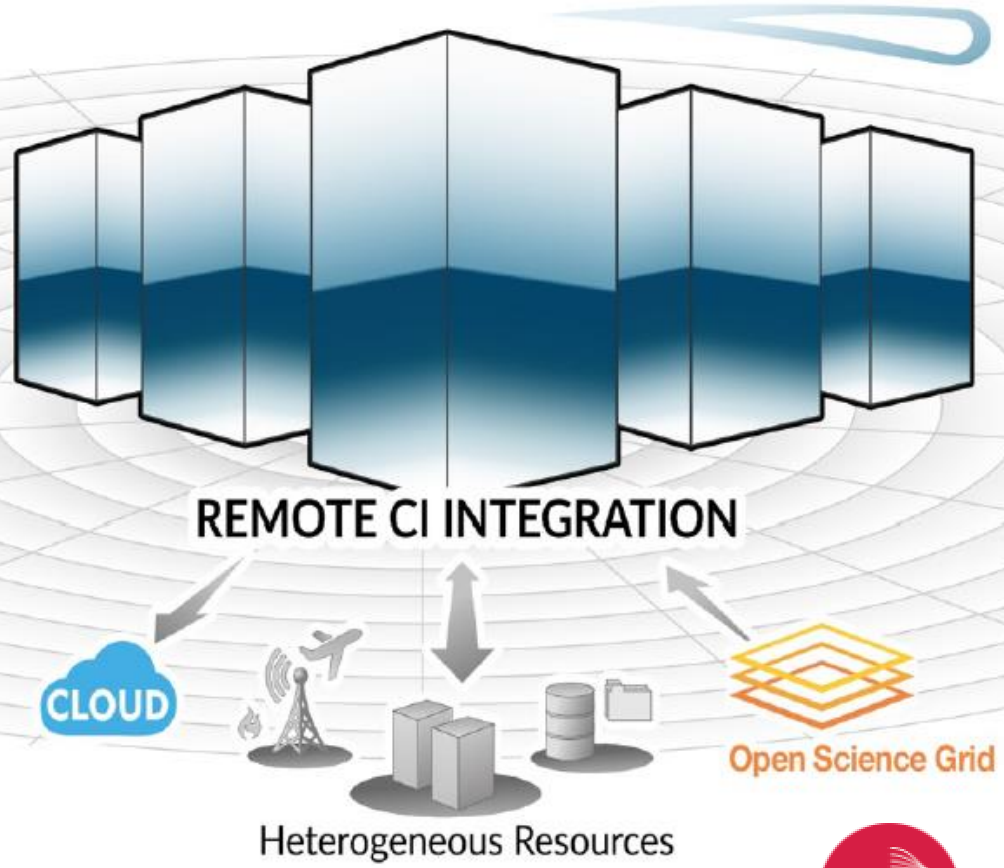
~~5~~ 7 PETAFLOP/S HPC and DATA RESOURCE

HPC RESOURCE

~~13~~ 17 Scalable Compute Units
728 Standard Compute Nodes
52 GPU Nodes: 208 GPUs
4 Large Memory Nodes

DATA CENTRIC ARCHITECTURE

12PB Perf. Storage: 140GB/s, 200k IOPS
Fast I/O Node-Local NVMe Storage
7PB Ceph Object Storage
High-Performance R&E Networking



LONG-TAIL SCIENCE

Multi-Messenger Astronomy
Genomics
Earth Science
Social Science

INNOVATIVE OPERATIONS

Composable Systems
High-Throughput Computing
Science Gateways
Interactive Computing
Containerized Computing
Cloud Bursting

PATh

NSF Award # 1928224

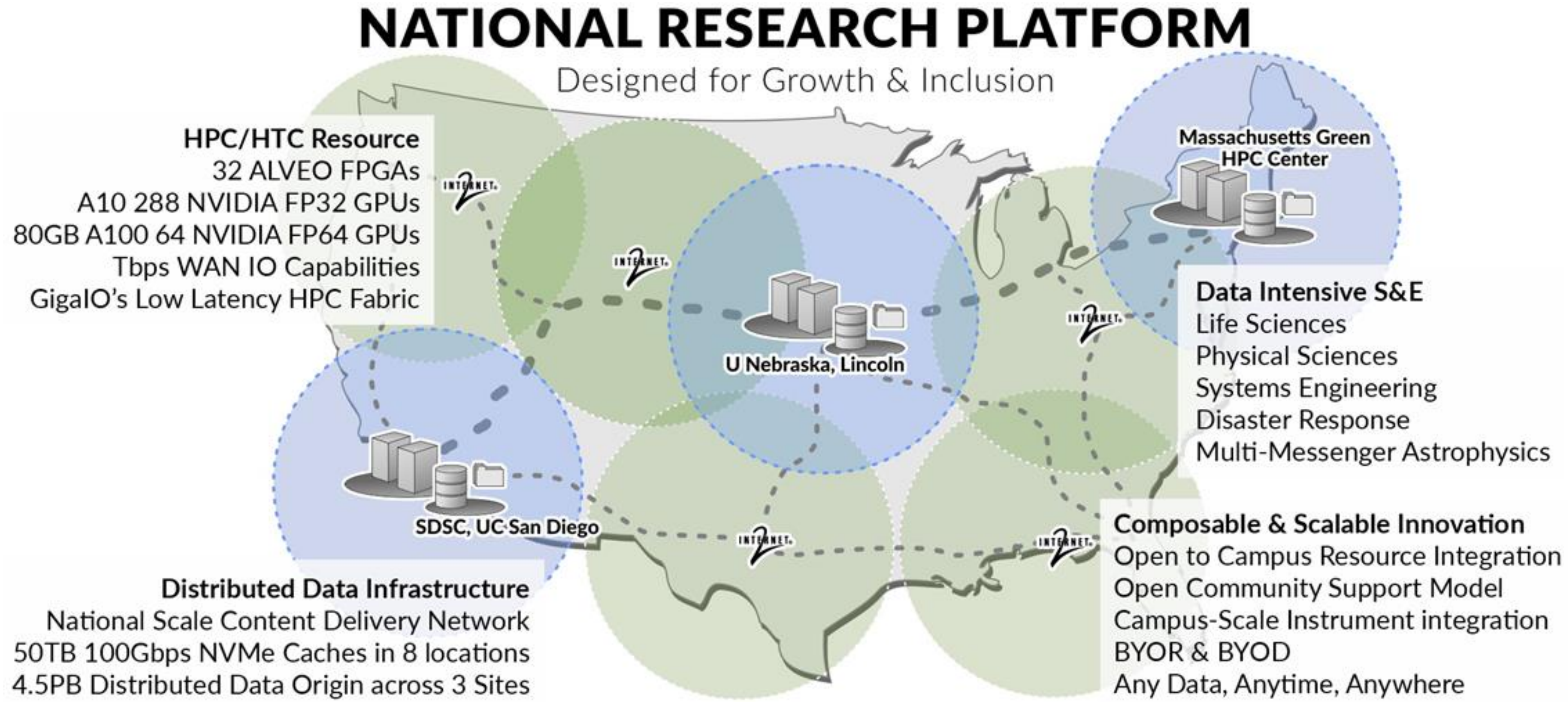
PIs: Mike Norman (PI), Ilkay Altintas, Amit Majumdar, Mahidhar Tatineni, Shawn Strande

Expanse Timeline

- ***Original Configuration (13 racks)**: Category 1: Capacity System, NSF Award # 1928224*
  - 728, 2-socket AMD Rome-based compute nodes (2.25 GHz EPYC; 64-core/socket). 93,184 compute cores in total. **Primarily ACCESS allocated. A fraction of this resource is available via NAIRR Pilot**
  - 52 4-way GPU nodes with V100 GPUs w/NVLINK
- ***Partnership to Advanced Throughput Computing (PATh) racks (2) – NSF funded, 2022:***
  - 112 2-socket AMD Milan-based compute nodes; 512 GB of memory per node
  - 8 4-way GPU nodes w/ A100 GPUs
- ***Industry rack (funded by UCSD/SDSC):***
  - 56 2-socket AMD Rome-based compute nodes
  - 4 4-way GPU nodes based on V100 w/NVLINK
- ***CW3E rack, 2023; Moved in 2025 to AWARE cluster***
  - 60 2-socket AMD Milan-based compute nodes; 256 GB of memory per node
- ***Expanse AI Resource (NAIRR Pilot supplement), 2025***
  - 34 Nodes: 2-socket Intel Sapphire Rapids CPUs, 1TB of memory, 4 H100 GPUs per node, 6.4TB NVMe
- **Highlighted resources can be allocated via NAIRR Pilot**

Expanse AI Resource (NAIRR)

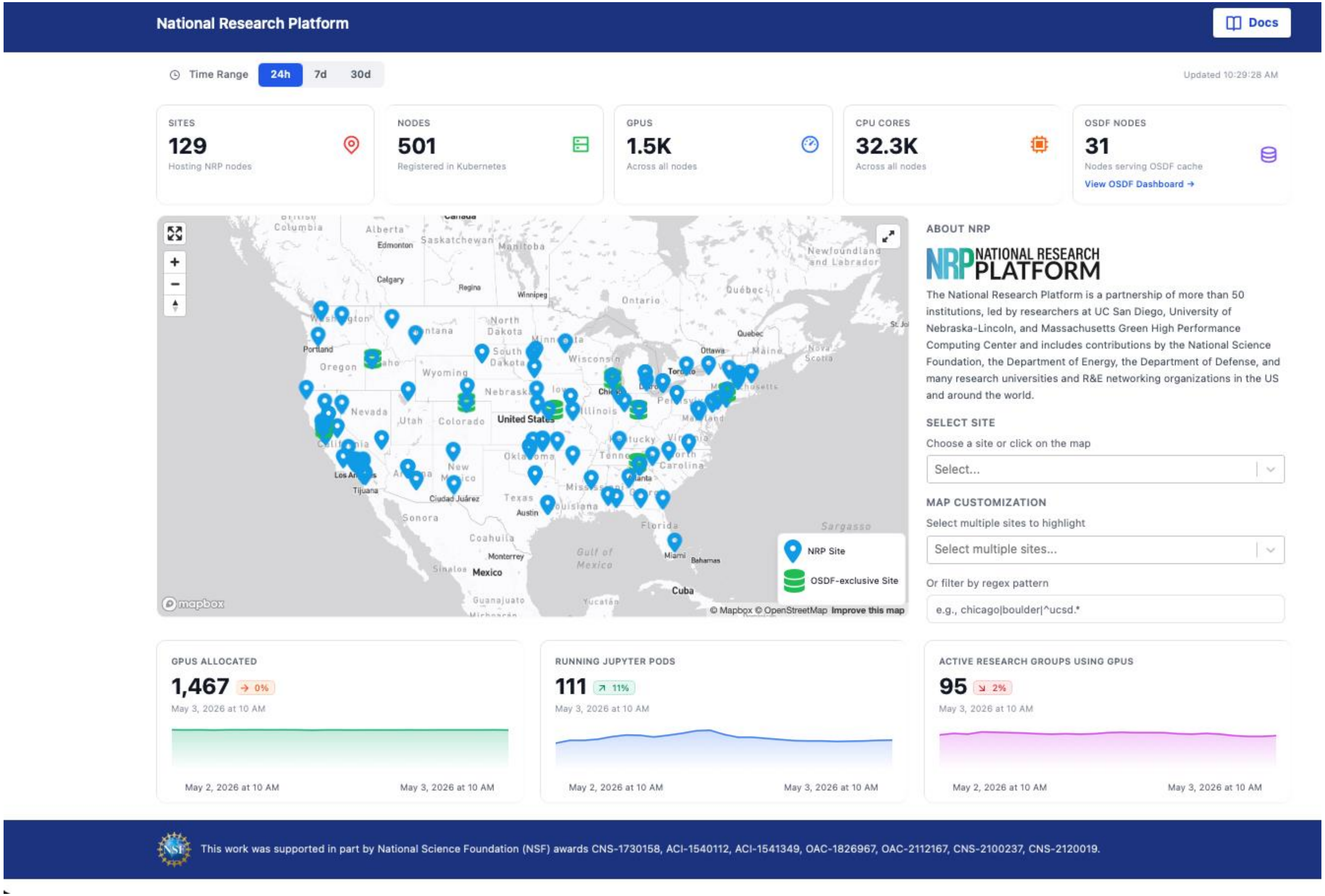
- New GPU computing and storage capacity recently integrated with *Expanse*
- Resources are available to the National AI Research Resource (NAIRR) Pilot project users
- Two SDSC Scalable Computing Units (SSCU) integrated by Dell with 17 GPU nodes per rack
- Each node has two 36-core Intel Sapphire Rapids CPUs, 1TB of memory, and 4 NVIDIA H100 GPUs; 6.4 TB of node-local NVMe;
- Networking uses NDR NVIDIA InfiniBand switches in the SSCUs. 2 NDR cards per node.
- NAIRR expansion will enable researchers to train complex models that can be applied to a wide range of real-life applications such as weather prediction, image classification, natural language processing, health care, software development, smart manufacturing and self driving cars.



Category II: A Prototype National Research Platform (PNRP) Project (NSF OAC #2112167) added significant compute and distributed storage resources to NRP

5-year project: $5M for Acquisition and Deployment; $7.25M for Operations and Maintenance

PI = Wuerthwein; Co-PIs: DeFanti, Rosing, Tatineni, Weitzel



https://dash.nrp-nautilus.io/

PNRP NAIRR Classroom: Resources

- Jupyter resources for classes that need AI/ML – can be teaching about AI/ML or using AI/ML in domain science
- Can use institutional authentication methods leveraging InCommon
- Professor and TA will have admin status to create namespaces for classes
- LLM inference services available, can be integrated into Jupyter
- Support/Community interaction through Matrix & Slack channels – one for the class that includes students so Professor/TA can help them and second for Jupyter admins that includes admins from other classes
- Direct ticketing system available for any issue needing longer follow ups
- GPU nodes each with 8 A10 GPUs, 512GB of RAM, 2 AMD EPYC 7502 CPUs, and 8TB of NVMe.
- Depending needs, other GPU resources can be provisioned via the NAIRR pilot and made available through this platform

PNRP Classroom: Proposal Requirements

- Syllabus of the planned class. PNRP can provide examples from previous classes.
- List of datasets that may be needed for the course
  - Open Data?
  - Location of data
  - Structure of data: #files, average size of file
- Number of Students
- Dates of classes
- Details of Jupyter notebook usage
  - Will they be used during class (if so the schedule)
  - Are quizzes/finals take home or in person (any special requirements)
- Typical resource requirements
  - #GPUs, memory, storage size

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Getting Access to NAIRR Resources!

- New to NAIRR? Get a startup to get familiar with the NAIRR resources:

https://nairrpilot.org/opportunities/startup-project

Enough resources to do modest projects. For larger projects you can use results from the startup to guide your submission for a research resource request

- If you have larger requests, use the NAIRR Research Resource opportunity:

https://nairrpilot.org/opportunities/allocations

- If you are an instructor needing time for students in your class to use:

https://nairrpilot.org/opportunities/education-call

Eligibility Requirements: Open to meritorious proposals by US-based researchers and educators from US-based institutions including academic institutions (including graduate students with a support letter from a faculty advisor), non-profits, federal agencies or federally funded R&D centers, state, local, or tribal agencies, startups and small businesses with federal grants.